

Phase 2 – Requirement Analysis

Project Title

Script-Controlled ACL – Restrict Record Access Based on Field Value

1. Introduction

Requirement Analysis is the process of identifying what the system needs to do before development begins. In this project, the requirements focus on creating a secure ServiceNow application that restricts access to student records based on user roles and the Branch field value.

2. Functional Requirements

The system must:

- Create a test user (EEE User).
- Create custom roles (bb1, bb2, bb3, bb4).
- Create the Institution Details custom table (u_institution_details).
- Add fields such as Student Roll Number, Student Name, Faculty Name, Branch, Email, Phone Number, and Description.
- Create multiple records for ECE, EEE, and CSE branches.
- Configure Read, Create, Write, and Delete ACLs.
- Support impersonation testing.

3. Non-Functional Requirements

- Security: Only authorized users should access records.
- Performance: ACL checks should execute quickly.
- Reliability: Access permissions should work consistently.
- Usability: The application should be easy to use.

4. User Requirements

User	Requirement
Admin	Full access
bb1	View EEE records
bb2	Create records
bb3	Edit records
bb4	Delete records

5. System Requirements

- ServiceNow Developer Instance
- Admin access
- Web browser
- Internet connection

6. Table Requirements

Table Name: u_institution_details

Fields

Field	Type
Student Roll Number	Auto Number
Student Name	Reference (User)
Faculty Name	Reference (User)
Branch	Choice
Email	String
Phone Number	String
Description	Multi-line String

7. Security Requirements

The Read ACL should verify:

- Admin users receive full access.
- Users with the bb1 role can access only EEE records.
- Other users cannot view restricted records.

Create, Write, and Delete permissions should be controlled using bb2, bb3, and bb4 roles.

8. Expected Output

The system should successfully protect student records using role-based security while allowing authorized users to perform only their permitted operations.